

DACNet: Distilled and Aligned Cyclemap Network for Efficient Time Series Forecasting

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Abstract—Leveraging global context from the full historical training data helps capture long-term patterns, thereby improving time series forecasting performance. However, directly modeling long sequences introduces substantial noise and redundancy, often leading to overfitting and high computational cost. To address these challenges, we present DACNet, a lightweight and efficient time series forecasting model that effectively exploits global information. We first propose a novel cycle wrap operation with overlap padding to transform the 1D historical series from the training set into a 2D cyclemap. Recognizing the redundancy within the cyclemap, we use two lightweight networks to perform inter-cycle compression and intra-cycle denoising, yielding a distilled global representation. Subsequently, for any given input instance, we obtain local features via a linear encoder and extract highly relevant global features from the distilled cyclemap through a cycle align operation and a similarity-based mechanism. Finally, we fuse these multi-scale features for prediction. Extensive experiments demonstrate that our approach achieves state-of-the-art performance across multiple real-world datasets while maintaining an extremely lightweight model with only tens of thousands of parameters.

Index Terms—time series forecasting, global information, temporal data mining, periodic pattern mining

I. INTRODUCTION

Time series forecasting plays a pivotal role in critical sectors such as intelligent transportation systems [1], financial markets [2], and renewable energy plants [3]. With technological advancement, the proposed methods have evolved from traditional statistical models [4], [5] to sophisticated deep learning architectures [6]–[8]. These models focus on capturing temporal dependencies within these local segments to project future temporal dynamics.

Fundamentally, the predictability of time series hinges on capturing robust intra-cycle patterns and inter-cycle evolutionary trends. However, the sample-independent learning paradigm, which treats samples in isolation, encounters an architectural limitation: the receptive field bottleneck [9]. They tend to erase the relative magnitude shifts between cycles, making it difficult to learn inter-cycle laws. Some recent methods [10], [11] attempt to leverage Global Information (GlobInfo) from the entire historical series, partly alleviating this limitation by expanding the field of view. Despite achieving impressive performance, they introduce a new bottleneck: inefficiency. To eliminate redundant information to avoid overfitting, they employ retrieval or clustering techniques to distill vast global segments, incurring substantial computa-

tional demands. Therefore, the critical challenge is not merely accessing global information, but distilling it effectively and efficiently.

To address this challenge, we propose a **Restructured Cyclemap** mechanism, as illustrated in Fig. 2a. By folding the 1D history based on its inherent periodicity, this transformation explicitly maps the two core sources of predictability onto orthogonal dimensions: **1) intra-cycle** dimension that preserves continuous phase variations (capturing periodic patterns), and **2) inter-cycle** dimension that reveals the evolution across identical phases of consecutive cycles (capturing trends). Recognizing the inherent sparsity within this structure, we distill the raw map into a compact, noise-free embedding that retains essential variation features. Finally, by strictly aligning the current prediction sample with this distilled map based on temporal phase, our model effectively retrieves highly correlated historical patterns to guide precise forecasting.

Building upon this insight, we propose DACNet (**D**istilled and **A**ligned **C**yclemap **N**etwork). This lightweight architecture implements the aforementioned cyclemap construction and, crucially, integrates a dual-branch forecasting module. By organically fusing the distilled global patterns (retrieved from the cyclemap) with the local context (encoded from the current sample), DACNet synthesizes a comprehensive representation that captures multi-scale temporal dependencies for precise prediction. Our main contributions are summarized as follows:

- 1) We propose a novel GlobInfo Restructuring strategy that transforms 1D time series into a 2D cyclemap. This structure significantly exposes data redundancy, allowing for the effective trimming of noise while preserving essential periodic and trend information.
- 2) We develop DACNet, an efficient forecasting architecture that integrates distilled global patterns and local context to capture multi-scale temporal dependencies.
- 3) Extensive experiments across eight real-world datasets demonstrate that DACNet achieves state-of-the-art performance. Notably, as shown in Fig. 1, our model delivers superior accuracy with lower computational overhead.

II. RELATED WORK

Various deep learning based methods have been proposed, leading to significant advancements in time series forecasting, including MLP-based models [12]–[15], RNN-based models

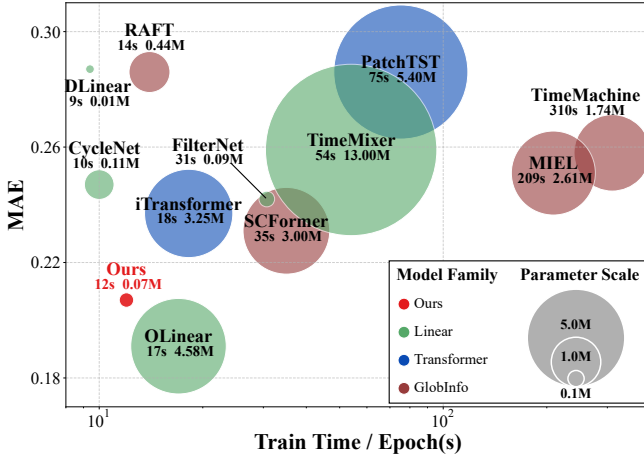


Fig. 1. Computational efficiency on the Solar dataset (The prediction horizon = 96. The look-back length = 96)

[16]–[19], CNN-based models [20]–[22], and Transformer-based models [23]–[26].

Some studies also investigate the efficiency of pretrained Large Language Models (LLMs) for time series analysis tasks, such as LLMTime [27], TEMPO [28], OFA [29], and PromptCast [30].

Methods of using Periodic Information: Many studies have recognized the importance of periodic information in time series forecasting and commonly adopt Seasonal-Trend Decomposition (STD) or Fourier Transform (FT) techniques. STD is used to effectively decouple periodicity and trend from the original series, such as DLinear [31], FEDformer [32], and Leddam [33]. The Sparse technique in SparseTSF [34] and RCF in CycleNet [35] are relatively novel variants of STD that achieve strong performance. Various specific types of FT are primarily employed to identify key components in the frequency domain, representing a more efficient expression of periodic patterns, such as adaptive frequency filters in FilterNet [36], complex-valued linear layers in FITS [37], frequency-domain MLPs in FreTS [38], and frequency debiased attention mechanisms in Fredformer [39]. In addition, recent studies have also proposed some other relatively novel approaches. For instance, DMMW [40] transforms series into images and leverages LVM to capture periodic patterns and trend variations, while TimeEmb [41] introduces a learnable time-invariant embedding bank to extract static patterns.

Methods of using Global Information: Within the traditional methodological framework, global information from the entire historical series is implicitly encoded within model parameters rather than being explicitly utilized. To break this bottleneck, recent advancements have focused on explicitly extracting global information [42]–[44]. TimeMachine [45] leverages the Mamba architecture to capture long-term dependencies from extended sequences. Retrieval-based approaches, such as RAFT [46], introduce retrieval-augmented mechanisms to fetch relevant historical segments directly. Similarly, CometNet [9] uses contextual motifs to guide forecasting,

while ReCast [11] employs a reliability-aware codebook to assist in modeling repetitive patterns.

However, these methods typically face trade-offs between performance and efficiency. Complex retrieval or memory mechanisms often incur high computational overheads and may introduce noise from redundant global information [47]. In contrast, DACNet achieves extreme efficiency by distilling restructured global temporal patterns.

III. METHODOLOGY

A. Preliminary

Problem Definition: As the proposed DACNet model adopts a **Channel-Independent (CI)** strategy, we simplify the notation for the remainder of this paper by focusing on the univariate case. Let $\mathcal{X} = \{x_1, \dots, x_N\} \in \mathbb{R}^N$ denote the entire time series, where N is the total number of time steps. Given a look-back window of length L , the input sample is denoted as $\mathbf{x}_t = \{x_t, \dots, x_{t+L-1}\} \in \mathbb{R}^L$. The goal of time series forecasting is to predict a future sequence of length T , denoted as $\mathbf{y}_t = \{x_{t+L}, \dots, x_{t+L+T-1}\} \in \mathbb{R}^T$. The objective is to learn a mapping function $f(\cdot)$ that minimizes the discrepancy between the prediction $\hat{\mathbf{y}}_t$ and the ground truth $\mathbf{y}_t$.

Forecasting with GlobInfo: Traditional forecasting models typically rely solely on local context within the look-back window, formulated as:

$$\hat{\mathbf{y}}_t = f(\mathbf{x}_t; \theta), \quad (1)$$

where θ denotes the learnable parameters.

In contrast, our approach introduces a global perspective by explicitly incorporating the entire historical training series, denoted as $\mathcal{X}_{\text{glob}} \in \mathbb{R}^{N_{\text{train}}}$, as an auxiliary static context. The proposed paradigm is formulated as:

$$\hat{\mathbf{y}}_t = f(\mathbf{x}_t, \mathcal{X}_{\text{glob}}; \theta). \quad (2)$$

Here, $\mathcal{X}_{\text{glob}}$ is transformed into a structured representation (i.e., the restructured cyclemap, denoted as $\mathbf{M}$) to provide global temporal patterns for alignment with the local input $\mathbf{x}_t$.

B. Structure Overview

As illustrated in Fig. 2, DACNet features a streamlined and efficient architecture primarily composed of lightweight MLP-based blocks. The framework comprises two phases: (1) **Cyclemap Construction**, which transforms the entire historical series into a structured, distilled representation; and (2) **Representation Fusion & Forecasting**, which fuses global patterns with local context for prediction.

Phase 1: Cyclemap Construction. The core innovation lies in restructuring the 1D entire history $\mathcal{X}_{\text{glob}}$ into a 2D matrix. To mitigate redundancy, we apply a Distillation mechanism (comprising compression and denoising operations) to yield a distilled static cyclemap, denoted as $\mathbf{M}^{(\text{st})}$. Subsequently, based on the temporal index of the current input sample $\mathbf{x}_t$, a specific segment is sliced from $\mathbf{M}^{(\text{st})}$ to generate the sample-aligned Dynamic Cyclemap, denoted as $\mathbf{M}_t^{(\text{dy})}$.

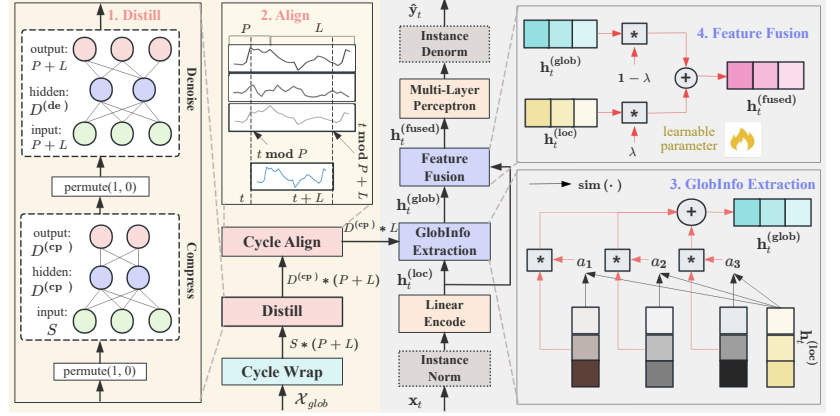
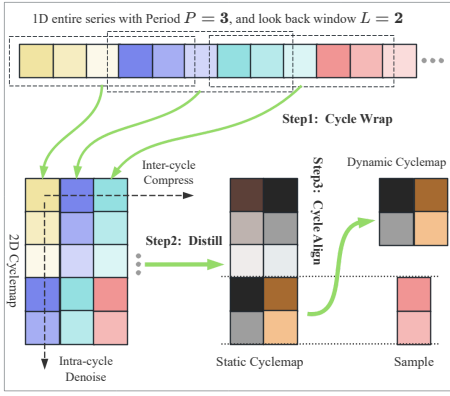


Fig. 2. Overview of DACNet. (a) Restructured cyclemap construction from the 1D entire historical series based on inherent periodicity. (b) DACNet architecture, consisting of Cyclemap Construction and Representation Fusion & Forecasting, which aligns distilled global patterns with local representations for prediction.

Phase 2: Representation Fusion & Forecasting. The forecasting process proceeds in four steps:

1) Local Encoding: The input sample $\mathbf{x}_t$ is projected via a linear encoder to extract local latent representations $\mathbf{h}_t^{(\text{loc})}$.

2) GlobInfo Extraction: The model computes the similarity between $\mathbf{h}_t^{(\text{loc})}$ and the periodic patterns within $\mathbf{M}_t^{(\text{dy})}$. Based on this mapping, relevant global features are retrieved via weighted aggregation to form $\mathbf{h}_t^{(\text{glob})}$.

3) Fusion: The local and global representations are adaptively fused to synthesize a comprehensive context vector.

4) Prediction: Finally, the fused representation is mapped to the future temporal horizon T via a Multi-Layer Perceptron (MLP) decoder to generate the forecast $\hat{\mathbf{y}}_t$.

The overall data flow can be mathematically formulated as follows:

$$\mathbf{M}_t^{(\text{dy})} = \text{construct}(\mathcal{X}_{\text{glob}}, t), \quad (3)$$

$$\mathbf{h}_t^{(\text{loc})} = \text{encode}(\mathbf{x}_t), \quad (4)$$

$$\mathbf{h}_t^{(\text{glob})} = \text{extract}(\mathbf{h}_t^{(\text{loc})}, \mathbf{M}_t^{(\text{dy})}), \quad (5)$$

$$\hat{\mathbf{y}}_t = \text{decode}(\mathbf{h}_t^{(\text{loc})} \oplus \mathbf{h}_t^{(\text{glob})}), \quad (6)$$

where $\oplus$ denotes the fusion operation (e.g., weighted addition with learnable parameters).

C. Cyclemap Construction

We propose a construction pipeline that transforms the 1D sequence into a structured 2D representation, effectively distilling task-relevant patterns while filtering out noise. This process consists of three stages: Cycle Wrap, Joint Compress and Denoise, and Cycle Align.

1) Cycle Wrap: To capture inherent periodic patterns, we first restructure the 1D history into a 2D matrix. Given the historical series $\mathcal{X}_{\text{glob}}$ of length N_{train} and a dominant period P , we employ a sliding window strategy with a kernel size of $(P + L)$ and a stride of P . Here, the stride P ensures cycle alignment, while the kernel extension L (equal to the sample's look-back window) creates an overlap to preserve boundary

continuity. This operation yields the Raw Cyclemap, denoted as $\mathbf{M}^{\text{raw}} \in \mathbb{R}^{S \times (P+L)}$, where S represents the total number of historical cycles (rows).

2) Joint Compress and Denoise: The raw cyclemap $\mathbf{M}^{\text{raw}}$ exhibits two distinct properties: (i) Along the inter-cycle dimension (rows), the series represents the evolution of trends, which typically vary slowly and contain redundant information; (ii) Along the intra-cycle dimension (columns), the series captures fine-grained periodic variations, often contaminated by high-frequency noise. Based on this, $\mathbf{M}^{\text{raw}}$ is a sparse matrix that can be effectively distilled. We propose a joint compression-denoising mechanism using lightweight MLP networks.

First, to reduce redundancy and computational cost, we apply Inter-cycle Compression. We project the high-dimensional cycle axis S into a compact latent space $D^{(\text{cp})}$ ($D^{(\text{cp})} \ll S$). This maps $\mathbf{M}^{\text{raw}}$ to an intermediate map $\mathbf{M}^{\text{tmp}} \in \mathbb{R}^{D^{(\text{cp})} \times (P+L)}$:

$$\mathbf{M}^{\text{tmp}} = \mathbf{W}_o^{(\text{cp})} \sigma \left(\mathbf{W}_i^{(\text{cp})} \mathbf{M}^{\text{raw}} + b^{(\text{cp})} \right), \quad (7)$$

where $\mathbf{W}_i^{(\text{cp})} \in \mathbb{R}^{D^{(\text{cp})} \times S}$ and $\mathbf{W}_o^{(\text{cp})} \in \mathbb{R}^{D^{(\text{cp})} \times D^{(\text{cp})}}$ are the compression weights.

Subsequently, to enhance signal clarity, we apply Intra-cycle Denoising to $\mathbf{M}^{\text{tmp}}$. We employ a bottleneck autoencoder structure along the feature dimension $(P + L)$ to filter out noise, resulting in the distilled **Static Cyclemap** $\mathbf{M}^{(\text{st})} \in \mathbb{R}^{D^{(\text{cp})} \times (P+L)}$:

$$\mathbf{M}^{(\text{st})} = \left(\sigma \left(\mathbf{M}^{\text{tmp}} \mathbf{W}_i^{(\text{de})} \right) + b^{(\text{de})} \right) \mathbf{W}_o^{(\text{de})}, \quad (8)$$

where $\mathbf{W}_i^{(\text{de})} \in \mathbb{R}^{(P+L) \times D^{(\text{de})}}$ and $\mathbf{W}_o^{(\text{de})} \in \mathbb{R}^{D^{(\text{de})} \times (P+L)}$ project the data into a lower-dimensional bottleneck and back, reconstructing the clean signal. Note that we purposely perform compression (7) before denoising (8). Since $D^{(\text{cp})} \ll S$, this order significantly reduces the computational complexity of the subsequent denoising step.

3) Cycle Align: Although $\mathbf{M}^{(st)}$ is distilled, it still contains phase-shifted information that may not align with the current input sample $\mathbf{x}_t$. To enforce temporal consistency, we perform a dynamic slicing operation. Let t be the timestamp of the first observation in the current sample $\mathbf{x}_t$. The relative phase index within a cycle is given by $\bar{t} = t \bmod P$. We intercept a segment of length L from $\mathbf{M}^{(st)}$ starting at index $\bar{t}$, generating the sample-specific Dynamic Cyclemap $\mathbf{M}_t^{(dy)} \in \mathbb{R}^{D^{(cp)} \times L}$:

$$\mathbf{M}_t^{(dy)} = \mathbf{M}^{(st)}[:, \bar{t} : \bar{t} + L]. \quad (9)$$

Through this process, $\mathbf{M}_t^{(dy)}$ retains the distilled global trend and periodic information while ensuring strict phase alignment with the local input sample.

D. Representation Fusion and Forecasting

Building upon the well-constructed Dynamic Cyclemap $\mathbf{M}_t^{(dy)}$, which distills global patterns synchronized with the current sample, we propose a fusion-based forecasting module. This module aligns global context with local features and decodes the synthesized representation for prediction. The process consists of three key stages: Local Encoding, Similarity-Based GlobInfo Extraction, and Adaptive Fusion & Decoding.

1) Local Encoding: We utilize a streamlined linear projection to extract local temporal features. Given the input sample $\mathbf{x}_t$, the local representation $\mathbf{h}_t^{(loc)} \in \mathbb{R}^L$ is obtained via:

$$\mathbf{h}_t^{(loc)} = W_e \mathbf{x}_t + b_e, \quad (10)$$

where $W_e \in \mathbb{R}^{L \times L}$ is the learnable weight matrix shared across channels. It functions effectively as a learnable filter, smoothing high-frequency noise in the raw sequence to facilitate subsequent matching.

2) GlobInfo Extraction: The core of DACNet is aligning the local query $\mathbf{h}_t^{(loc)}$ with the global keys provided by $\mathbf{M}_t^{(dy)}$. To measure the relevance between the current local context and these global prototypes, we employ a similarity function $\text{sim}(\cdot, \cdot)$ based on the Negative 1-Norm (Manhattan Distance), which is robust to outliers.

The attention weights $\mathbf{a} \in \mathbb{R}^{D^{(cp)}}$ are computed by normalizing the similarity scores:

$$\mathbf{a} = \text{Softmax} \left(\frac{1}{\sqrt{L}} \text{sim}(\mathbf{h}_t^{(loc)}, \mathbf{M}_t^{(dy)}) \right), \quad (11)$$

where the Softmax operation is applied along the dimension of global prototypes ($D^{(cp)}$).

Subsequently, the aligned global representation $\mathbf{h}^{(glob)} \in \mathbb{R}^L$ is retrieved via the weighted aggregation:

$$\mathbf{h}_t^{(glob)} = \mathbf{a} \mathbf{M}_t^{(dy)}. \quad (12)$$

Here, the vector-matrix multiplication aggregates the prototypes based on their relevance to the current local shape.

3) Adaptive Fusion and Decoding: To organically integrate local and global contexts, we design a learnable Gating Mechanism. Recognizing that the importance of global information varies across different time steps, we define a temporal gate $\lambda \in \mathbb{R}^L$:

$$\lambda = \sigma(\mathbf{w}_{\text{gate}}), \quad (13)$$

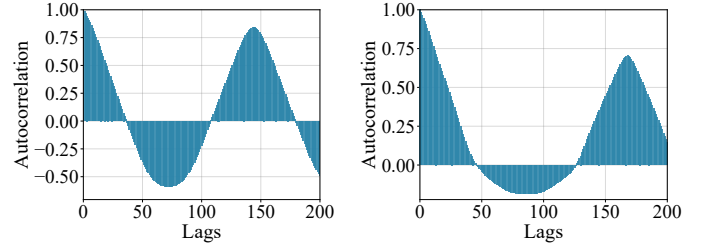


Fig. 3. Representative ACF profiles used for period estimation. The prominent peaks at non-zero lags reveal recurring temporal patterns and provide candidate cycle lengths for constructing the cyclemap.

where $\mathbf{w}_{\text{gate}} \in \mathbb{R}^L$ is a learnable parameter vector capturing temporal disparities, and $\sigma(\cdot)$ is the Sigmoid function.

The final fused representation $\mathbf{h}_t^{(\text{fused})}$ is obtained via adaptive interpolation:

$$\mathbf{h}_t^{(\text{fused})} = \lambda \odot \mathbf{h}_t^{(loc)} + (1 - \lambda) \odot \mathbf{h}_t^{(glob)}, \quad (14)$$

where $\odot$ denotes the element-wise Hadamard product.

Finally, a MLP decoder maps the fused representation to the future horizon T :

$$\hat{\mathbf{y}} = \text{MLP}(\mathbf{h}_t^{(\text{fused})}). \quad (15)$$

This architecture highlights a key insight: when effective global patterns are explicitly retrieved, a simple MLP decoder suffices to achieve state-of-the-art performance, suggesting that supplementing local context with external global information may be as crucial as modeling complex local dynamics.

IV. EXPERIMENTS

We conduct experiments on eight widely used real-world benchmarks, including ETT (ETTh1, ETTh2, ETTm1, ETTm2), Electricity, Traffic, Weather, and Solar. These datasets span diverse domains and exhibit distinct characteristics in terms of scale, dimensionality, and periodicity, ensuring a robust and general evaluation. Following prior work such as CycleNet [35], the dominant period of each dataset can be estimated using the autocorrelation function (ACF), where a prominent peak at a non-zero lag indicates a recurring temporal pattern. Fig. 3 presents Solar and Electricity as representative examples, whose pronounced secondary ACF peaks provide a clear basis for determining their cycle lengths.

A. Main Results

1) Baselines and Setup: We compare DACNet with nine representative forecasting baselines: TimeMachine [45] and RAFT [46] as recent global-information or retrieval-oriented methods; DLinear [31], CycleNet [35], and TimeMixer [49] as linear, cycle-aware, or mixing-based methods; and iTransformer [7], TimeMosaic [50], PatchTST [6], and TQNet [51] as Transformer-, patch-, or query-based methods. The baseline results are taken from their respective original papers.

Following the same data partitioning protocol as the baselines, the four ETT datasets are split into training, validation, and test sets at a ratio of 2 : 1 : 1, while the remaining datasets

TABLE I
MULTIVARIATE TIME SERIES FORECASTING RESULTS WITH LOOK-BACK WINDOW $L = 96$ AND PREDICTION LENGTHS $T \in \{96, 192, 336, 720\}$. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND BEST ARE UNDERLINED.

Dataset	Len	DACNet (Ours)	TimeMachine (2024)	RAFT (2025)	CycleNet (2024a)	TimeMixer (2024)	iTransformer (2023a)	TimeMosaic (2025)	PatchTST (2023)	TQNet (2025a)	DLinear (2023)
		MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE
ETTm1	96	0.297 0.333	0.317 0.355	0.329 0.371	0.319 0.36	0.32 0.357	0.334 0.368	<u>0.308 0.338</u>	0.329 0.367	0.311 0.353	0.345 0.372
	192	0.35 0.361	0.357 0.378	0.363 0.388	0.36 0.381	0.361 0.381	0.377 0.391	<u>0.364 0.368</u>	0.367 0.385	<u>0.356 0.378</u>	0.38 0.389
	336	0.375 0.383	0.379 0.399	0.391 0.408	0.389 0.403	0.39 0.404	0.426 0.42	<u>0.394 0.39</u>	0.399 0.41	0.39 0.401	0.413 0.413
	720	0.431 0.418	<u>0.445</u> 0.436	<u>0.444</u> 0.438	0.447 0.441	0.454 0.441	0.491 0.459	0.46 <u>0.428</u>	0.454 0.439	0.452 0.44	0.474 0.453
ETTm2	96	0.163 0.241	0.175 0.256	0.177 0.265	0.163 0.246	0.175 0.258	0.18 0.264	<u>0.17 0.248</u>	0.175 0.259	0.173 0.256	0.193 0.292
	192	0.227 0.283	0.239 0.299	0.241 0.307	<u>0.229 0.29</u>	0.237 0.299	0.25 0.309	0.236 0.291	0.241 0.302	0.238 0.298	0.284 0.362
	336	0.284 0.323	<u>0.287</u> 0.332	0.302 0.346	0.284 0.322	0.298 0.34	0.311 0.348	0.295 0.331	0.305 0.343	0.301 0.34	0.369 0.427
	720	<u>0.381 0.383</u>	0.371 <u>0.385</u>	0.401 0.403	0.389 0.391	0.391 0.396	0.412 0.407	0.392 0.386	0.402 0.4	0.397 0.396	0.554 0.522
ETTh1	96	<u>0.366 0.382</u>	0.364 <u>0.387</u>	0.387 0.414	0.375 0.395	0.375 0.4	0.386 0.405	0.369 0.389	0.414 0.419	0.371 0.393	0.386 0.4
	192	0.42 0.41	0.415 0.416	0.423 0.431	0.436 0.428	0.429 0.421	0.441 0.436	<u>0.416 0.415</u>	0.46 0.445	0.428 0.426	0.437 0.432
	336	0.455 <u>0.429</u>	0.429 0.421	0.456 0.445	0.496 0.455	0.484 0.458	0.487 0.458	<u>0.454 0.435</u>	0.501 0.466	0.476 0.446	0.481 0.459
	720	0.463 <u>0.447</u>	0.458 <u>0.453</u>	0.463 0.464	0.52 0.484	0.498 0.482	0.503 0.491	<u>0.461 0.458</u>	0.5 0.488	0.487 0.47	0.519 0.516
ETTh2	96	0.275 0.326	0.275 0.334	0.287 0.34	0.298 0.344	0.289 0.341	0.297 0.349	<u>0.286 0.332</u>	0.292 0.342	0.295 0.343	0.333 0.387
	192	<u>0.355 0.376</u>	0.349 0.381	0.383 0.401	0.372 0.396	0.372 0.392	0.38 0.4	0.362 <u>0.38</u>	0.387 0.4	0.367 0.393	0.477 0.476
	336	0.401 0.415	0.34 0.381	0.431 0.445	0.431 0.439	<u>0.386 0.414</u>	0.428 0.432	0.409 0.419	0.426 0.433	0.417 0.427	0.594 0.541
	720	<u>0.41 0.429</u>	0.411 0.433	0.445 0.466	0.45 0.458	0.412 0.434	0.427 0.445	0.396 0.422	0.431 0.446	0.433 0.446	0.831 0.657
Electricity	96	<u>0.135 0.224</u>	0.142 0.236	0.165 0.266	<u>0.136 0.229</u>	0.153 0.247	0.148 0.24	0.16 0.258	0.161 0.25	0.134 0.229	0.197 0.282
	192	<u>0.153 0.24</u>	0.158 0.25	0.169 0.268	0.152 0.244	0.166 0.256	0.162 0.253	0.174 0.269	0.199 0.289	0.154 0.247	0.196 0.285
	336	0.168 0.257	0.178 0.268	0.18 0.279	0.17 <u>0.264</u>	0.185 0.277	0.178 0.269	0.189 0.283	0.215 0.305	<u>0.169 0.264</u>	0.209 0.301
	720	<u>0.206 0.291</u>	0.207 0.298	0.21 <u>0.292</u>	0.212 0.299	0.225 0.31	0.225 0.317	0.225 0.304	0.256 0.337	0.201 0.294	0.245 0.333
Traffic	96	0.402 0.246	0.397 0.268	0.516 0.336	0.458 0.296	0.462 0.285	0.395 0.268	0.424 0.269	0.446 0.283	0.413 <u>0.261</u>	0.65 0.396
	192	<u>0.425 0.258</u>	0.417 0.274	0.497 0.322	0.457 0.294	0.473 0.296	0.417 0.276	0.448 0.279	0.54 0.354	0.432 <u>0.273</u>	0.598 0.37
	336	<u>0.44 0.266</u>	0.433 0.281	0.505 0.323	0.47 0.299	0.498 0.296	0.433 0.283	0.464 0.284	0.551 0.358	0.45 <u>0.277</u>	0.605 0.373
	720	<u>0.468 0.28</u>	0.467 0.3	0.502 0.33	0.502 0.314	0.506 0.313	0.467 0.302	0.495 0.301	0.586 0.375	0.486 <u>0.295</u>	0.645 0.394
Weather	96	0.153 0.191	0.164 0.208	0.191 0.24	0.158 0.203	0.163 0.209	0.174 0.214	0.165 <u>0.198</u>	0.177 0.218	<u>0.157</u> 0.2	0.196 0.255
	192	0.2 0.236	0.211 0.25	0.241 0.284	0.207 0.247	0.208 0.25	0.221 0.254	0.215 <u>0.243</u>	0.225 0.259	<u>0.206 0.245</u>	0.237 0.296
	336	0.258 0.28	0.256 0.29	0.295 0.326	0.262 0.289	0.251 0.287	0.278 0.296	0.272 <u>0.286</u>	0.278 0.297	0.262 0.287	0.283 0.335
	720	0.338 0.334	<u>0.342</u> 0.343	0.372 0.377	0.344 0.344	<u>0.339</u> 0.341	0.358 0.347	0.353 <u>0.34</u>	0.354 0.348	0.344 0.342	0.345 0.381
Solar	96	<u>0.18 0.207</u>	0.216 0.258	0.225 0.286	0.19 0.247	0.189 0.259	0.203 0.237	0.209 0.248	0.234 0.286	0.173 0.233	0.29 0.378
	192	0.196 0.218	0.242 0.271	0.247 0.31	0.21 0.266	0.222 0.283	0.233 0.261	0.238 0.272	0.267 0.31	<u>0.199 0.257</u>	0.32 0.398
	336	0.203 0.225	0.264 0.288	0.265 0.336	0.217 0.266	0.231 0.292	0.248 0.273	0.255 0.277	0.29 0.315	<u>0.211 0.263</u>	0.353 0.415
	720	<u>0.21 0.23</u>	0.255 0.28	0.259 0.338	0.223 <u>0.266</u>	0.223 0.285	0.249 0.275	0.257 0.284	0.289 0.317	0.209 0.27	0.356 0.413

use a ratio of 7 : 1 : 2. We adopt a fixed look-back window $L = 96$ and four prediction horizons $T \in \{96, 192, 336, 720\}$. DACNet is trained for at most 30 epochs with a batch size of 32 and early stopping patience of 3. We fix the denoising dimension to $D^{(de)} = 16$ and search the learning rate over $\{0.01, 0.005, 0.001\}$, the compression dimension $D^{(cp)}$ over $\{8, 16, 32\}$, and the MLP hidden dimension $D^{(mlp)}$ over $\{128, 256, 512, 1024\}$; the best-performing configuration is reported for each setting. The model is implemented in PyTorch and executed on an NVIDIA RTX A6000 GPU. Performance is evaluated using Mean Squared Error (MSE) and Mean Absolute Error (MAE), where lower values indicate better forecasting accuracy. The source code is available at: <https://github.com/yuliang555/DACNet>.

2) Results: Table I reports the horizon-level forecasting results. Across the 64 dataset-horizon-metric comparisons, DACNet obtains the best or tied-best value in 43 cases and remains within the top two in 57 cases. Its performance is most stable on ETTm1, ETTm2, Weather, and Solar, where it usually provides the lowest or second-lowest error across horizons, especially on MAE. The advantage is less uniform

on ETTh and Traffic: TimeMachine obtains lower MSE in several ETTh settings, iTransformer gives the lowest MSE on Traffic at $T = 96$, and TimeMixer gives the lowest MSE on Weather at $T = 336$. These exceptions indicate that DACNet provides broad accuracy gains, but the relative benefit still depends on the dataset, horizon, and metric.

B. Ablation Study

1) Global and Local Features: To quantify the contribution of each scale, we conduct ablation studies by freezing the gating parameter λ (no longer learnable):

- 1) Local-Only:** Setting $\lambda = 1$ to mask global information;
- 2) Global-Only:** Setting $\lambda = 0$ to disregard local information.

As illustrated in Fig. 4, removing information from either scale leads to a consistent performance degradation across all datasets, validating the necessity of the dual-scale fusion mechanism. The results reveal distinct patterns based on dataset characteristics:

- **Low-Sensitivity Group:** For ETTm1, ETTm2, and Weather, the performance drop is relatively minor ($\Delta\text{MAE} < 0.15$), and the impact of removing global vs. local features

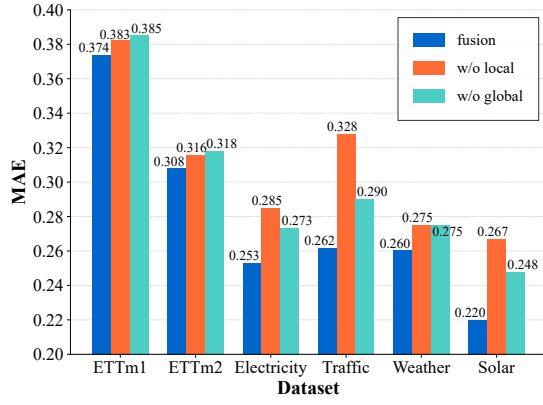


Fig. 4. Ablation study on global and local feature fusion. We compare the full model (fusion) against variants without local information ($\lambda = 0.0$, w/o local) or without global information ($\lambda = 1.0$, w/o global).

is comparable. This suggests these datasets possess relatively stable periodic patterns that are partially redundant across scales.

- **High-Sensitivity Group:** For Electricity, Traffic, and Solar, the degradation is substantial (up to 0.06 MAE), particularly for Traffic. The significant gap between the two ablation settings indicates that these datasets contain complex dependencies where global trends and local variations carry non-overlapping, complementary information.

Furthermore, an emerging trend suggests that as dataset dimensionality increases (e.g., Traffic), the model becomes increasingly reliant on the fusion mechanism, with local features showing a slightly higher critical weight in high-dimensional scenarios.

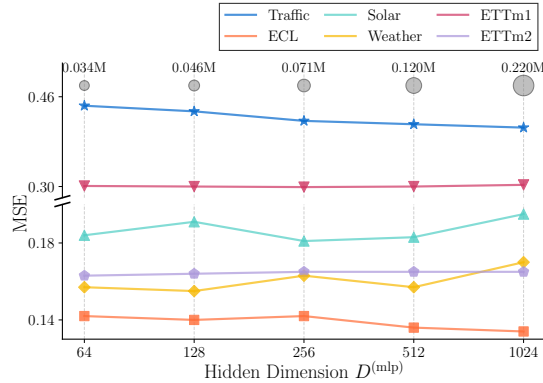


Fig. 5. Impact of hidden dimension $D^{(mlp)}$ on MSE performance across six datasets.

2) *Cyclemap compression and denoising:* The Distillation Module is designed to filter redundancy and noise from the sparse 2D cyclemap. To validate its components, we evaluate three variants: Compression-only, Denoising-only, and the proposed Joint (Compression + Denoising) mechanism. As shown in Table II, the Joint operations consistently achieve the optimal performance across all six benchmark datasets, proving its capability to enhance model generalization.

- **Noise Robustness:** The improvement is most pronounced on the Traffic dataset (MSE reduction ≈ 0.01). Notably, applying Compression alone on Traffic degrades performance, likely because compression forces the model to retain high-amplitude outliers (e.g., congestion spikes) as “features.” The Denoising step is thus critical for filtering such sporadic noise.

- **General Enhancement:** For datasets like ETT and Weather, the Joint module provides stable, albeit subtle, improvements, confirming that simultaneously reducing redundancy (Inter-cycle) and removing noise (Intra-cycle) is the most effective strategy for global feature extraction.

C. Efficiency Analysis

The efficiency of DACNet is a core design objective, achieved through its strictly linear backbone and the strategic dimensionality reduction in the cyclemap. We provide a comprehensive analysis of the model’s complexity from two perspectives: parameter count (Space Complexity) and computational speed (Time Complexity).

1) **Parameter Analysis:** DACNet maintains an ultra-lightweight structure. The total parameter count can be decomposed into two primary components:

- **Feature Extraction Modules:** This includes the Linear Encoder (L^2), Distillation weights (approximated as $S \cdot D^{(cp)} + (P + L) \cdot D^{(de)}$), and the gating vector $\mathbf{w}_{gate}$ (L). Since we design the latent dimensions $D^{(cp)}$ and $D^{(de)}$ to be small, the parameters contributed by these modules are minimal.

- **Prediction Decoder:** The MLP decoder dominates the parameter count, quantified as $D^{(mlp)} \cdot (L + T)$, where $D^{(mlp)}$ is the hidden dimension.

As illustrated in Fig. 5, we investigate the sensitivity of model performance to the hidden dimension $D^{(mlp)}$. In most scenarios (e.g., ETT and Weather), the model saturates with $D^{(mlp)} \leq 256$, keeping the total parameter count at the 10k level. While larger-scale datasets like Electricity and Traffic benefit from a wider hidden layer (requiring slightly more parameters), the overall size remains orders of magnitude smaller than Transformer-based baselines (typically millions of parameters), as evidenced in Fig. 1. Excessively increasing $D^{(mlp)}$ yields diminishing returns and may lead to overfitting.

2) **Computational Speedup:** Beyond parameter efficiency, the *Distillation* module plays a pivotal role in accelerating training. By compressing the raw cyclemap along the inter-cycle dimension, we reduce the computational complexity of the subsequent operations by an order of magnitude. To quantify this benefit, we conducted speed tests on Electricity, Solar, and Traffic, which possess varying historical lengths (resulting in raw inter-cycle dimensions S of 109, 254, and 72, respectively).

Fig. 6 presents the training speed comparison with and without compression. The results demonstrate that:

- **Minimal Information Loss:** The compression operation results in marginal performance fluctuations, confirming that the discarded information is indeed redundant.

- **Significant Acceleration:** Compression yields substantial speedups. Notably, the acceleration gain is positively corre-

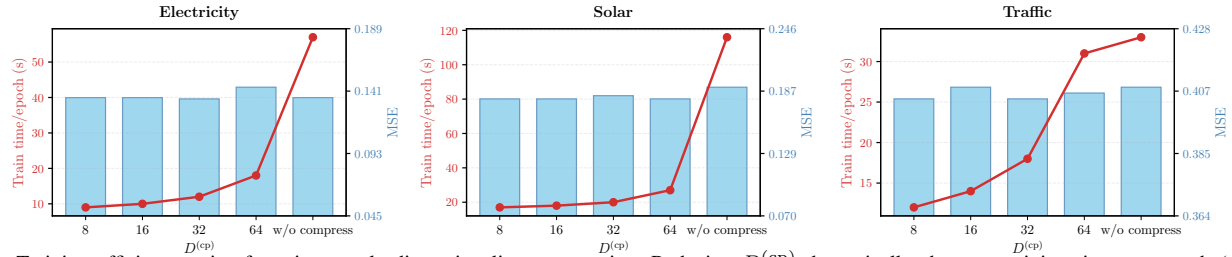


Fig. 6. Training efficiency gains from inter-cycle dimensionality compression. Reducing $D^{(cp)}$ dramatically shortens training time per epoch (red lines) across all datasets, while incurring negligible impact on forecasting accuracy, as indicated by the stable MSE values (blue bars).

TABLE II

ABLATION STUDY ON THE EFFECTS OF COMPRESSION AND DENOISING MODULES ACROSS SIX BENCHMARK DATASETS. WE EVALUATE FOUR CONFIGURATIONS WITH DIFFERENT COMBINATIONS OF MODULES. THE LOOK-BACK WINDOW IS FIXED AT $L = 96$, AND THE REPORTED RESULTS ARE AVERAGED OVER PREDICTION HORIZONS $T \in \{96, 192, 336, 720\}$. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Compress	Denoise	ETTm1		ETTm2		Electricity		Traffic		Weather		Solar	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
✓	✓	0.363	0.374	0.264	0.308	0.166	0.253	0.434	0.262	0.238	0.260	0.197	0.220
✓	✗	0.368	0.375	0.265	0.309	0.169	0.259	0.486	0.290	0.241	0.263	0.200	0.222
✗	✓	0.368	0.375	0.266	0.309	0.167	0.255	0.440	0.267	0.240	0.263	0.202	0.221
✗	✗	0.367	0.377	0.266	0.310	0.174	0.263	0.444	0.267	0.241	0.262	0.199	0.221

TABLE III

EFFECT OF USING THE CYCLEMAP AS A PLUG-IN MODULE FOR DIFFERENT FORECASTING BACKBONES ON ELECTRICITY AND TRAFFIC. “REDUCTION” DENOTES THE RELATIVE ERROR REDUCTION AFTER ADDING THE CYCLEMAP.

Model	Electricity								Traffic							
	$T = 96$		$T = 192$		$T = 336$		$T = 720$		$T = 96$		$T = 192$		$T = 336$		$T = 720$	
iTransformer	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
	0.148	0.240	0.162	0.253	0.178	0.269	0.225	0.317	0.395	0.268	0.417	0.276	0.433	0.283	0.467	0.302
	0.132	0.222	0.152	0.240	0.165	0.254	0.192	0.279	0.392	0.240	0.414	0.250	0.428	0.257	0.464	0.279
+ Cyclemap	10.8	7.5	6.2	5.1	7.3	5.6	14.7	12.0	0.8	10.4	0.7	9.4	1.2	9.2	0.6	7.6
MLP	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
	0.175	0.259	0.181	0.265	0.197	0.282	0.240	0.317	0.500	0.325	0.496	0.321	0.509	0.325	0.542	0.342
	0.134	0.223	0.153	0.240	0.168	0.257	0.206	0.291	0.402	0.246	0.425	0.258	0.440	0.266	0.468	0.280
+ Cyclemap	23.4	13.9	15.5	9.4	14.7	8.9	14.2	8.2	19.6	24.3	14.3	19.6	13.6	18.2	13.7	18.1
Linear	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
	0.197	0.274	0.197	0.277	0.212	0.292	0.253	0.324	0.645	0.383	0.598	0.361	0.605	0.362	0.643	0.381
	0.142	0.234	0.160	0.249	0.175	0.267	0.215	0.298	0.466	0.312	0.462	0.303	0.484	0.320	0.510	0.330
+ Cyclemap	27.9	14.6	18.8	10.1	17.5	8.6	15.0	8.0	27.8	18.5	22.7	16.1	20.0	11.6	20.7	13.4

lated with the length of the historical series. For the longest dataset, Solar, the training speed is accelerated by up to $7\times$, whereas for the shorter Traffic dataset, a $3\times$ improvement is observed.

This confirms that DACNet is not only lightweight in storage but also highly scalable to long-term historical data.

D. Generalization and Robustness Analysis

1) Cyclemap as a Plug-in Module: We integrate the cyclemap module into iTransformer, MLP, and Linear backbones and evaluate them on Electricity and Traffic. As shown in Table III, adding the cyclemap reduces both MSE and MAE for every backbone, horizon, and dataset reported. The gains are especially pronounced for the simpler MLP and Linear backbones, reaching 27.9% in MSE and 24.3% in MAE. For iTransformer on Traffic, the MSE reductions are comparatively small (0.6%–1.2%), whereas the MAE reductions remain between 7.6% and 10.4%. Thus, the cyclemap

is broadly compatible with different architectures, although its contribution depends on the backbone and metric.

2) Weakly Periodic Data: We further evaluate DACNet on Exchange, a financially driven dataset without an obvious stable period. We set $P = 1$, which avoids imposing a multi-step periodic prior and allows the model to operate on the long-term evolution across individual time steps. Table IV shows that DACNet achieves the lowest MSE and MAE at $T \in \{96, 192, 720\}$. At $T = 336$, PatchTST performs best, indicating that the $P = 1$ setting extends DACNet to weakly periodic data but does not guarantee uniform superiority at every horizon.

3) Period Sensitivity: We examine DACNet with candidate periods $P \in \{24, 96, 144, 168\}$ on ETTm1, Electricity, and Solar. As shown in Table V, the period identified for each dataset (96, 168, and 144, respectively) yields the lowest MSE and MAE at $T = 96$. ETTm1 changes only moderately under alternative periods, whereas Electricity shows a larger

TABLE IV
FORECASTING RESULTS ON THE WEAKLY PERIODIC EXCHANGE
DATASET WITH $P = 1$.

Horizon	DACNet		iTransformer		PatchTST	
	MSE	MAE	MSE	MAE	MSE	MAE
96	0.083	0.200	0.086	0.206	0.088	0.205
192	0.174	0.295	0.177	0.299	0.176	0.299
336	0.339	0.420	0.331	0.417	0.301	0.397
720	0.794	0.670	0.847	0.691	0.901	0.714

TABLE V
PERIOD SENSITIVITY OF DACNET AT $T = 96$. THE BEST RESULT FOR
EACH DATASET AND METRIC IS HIGHLIGHTED IN **BOLD**.

Dataset	$P = 24$		$P = 96$		$P = 144$		$P = 168$	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTM1	0.306	0.338	0.297	0.333	0.307	0.337	0.305	0.338
Electricity	0.162	0.245	0.163	0.245	0.163	0.246	0.135	0.224
Solar	0.194	0.226	0.189	0.216	0.180	0.207	0.198	0.225

degradation when its period is misspecified. These results indicate that accurate period selection is beneficial, although the degree of sensitivity varies across datasets.

4) Stability across Random Seeds: We repeat representative experiments with five random seeds while keeping the model configuration fixed. Table VI reports standard deviations between 0.001 and 0.007. The variation is small relative to the corresponding mean errors, suggesting that the reported performance is not driven by a particular initialization.

TABLE VI
SEED STABILITY OF REPRESENTATIVE DACNET MSE RESULTS UNDER A
FIXED CONFIGURATION.

Dataset	2024	2025	2026	2027	2028	Mean $\pm$ Std.
Solar	0.210	0.223	0.213	0.213	0.216	0.217 $\pm$ 0.007
Weather	0.342	0.341	0.342	0.343	0.343	0.342 $\pm$ 0.001
Traffic	0.472	0.471	0.471	0.471	0.471	0.471 $\pm$ 0.001
Electricity	0.204	0.206	0.211	0.207	0.206	0.208 $\pm$ 0.003
ETTM1	0.437	0.435	0.435	0.437	0.433	0.435 $\pm$ 0.002

5) Comparison with SVD: We compare the proposed MLP-based distillation with rank-8 singular value decomposition (SVD) on Electricity. As reported in Table VII, SVD improves over using no compression or denoising, while the proposed method obtains the lowest MSE and MAE at both $T = 96$ and $T = 720$. Relative to SVD, the improvements are larger at $T = 96$ and remain modest but consistent at $T = 720$, supporting the use of a learnable nonlinear distillation operator.

6) Channel Mixing for Cross-Variate Dependencies: We explore an optional Channel Mixing extension, which introduces a mixing operation sandwiched between the compression and denoising stages to capture cross-variate dependencies. This module functions as a “plug-and-play” extension for DACNet, specifically tailored for datasets exhibiting strong inter-channel dependencies. As shown in Figure 7, the integration of the Channel Mixing plug-in yields a marked

TABLE VII
COMPARISON OF CYCLEMAP COMPRESSION AND DENOISING
STRATEGIES ON ELECTRICITY.

Method	$T = 96$		$T = 720$	
	MSE	MAE	MSE	MAE
None	0.147	0.237	0.213	0.299
SVD (rank 8)	0.142	0.232	0.210	0.295
MLP (ours)	0.135	0.224	0.206	0.291

performance improvement. This confirms that while Channel-Independence is efficient for general cases, explicitly modeling cross-channel interactions via the cyclemap is beneficial for spatially correlated data.

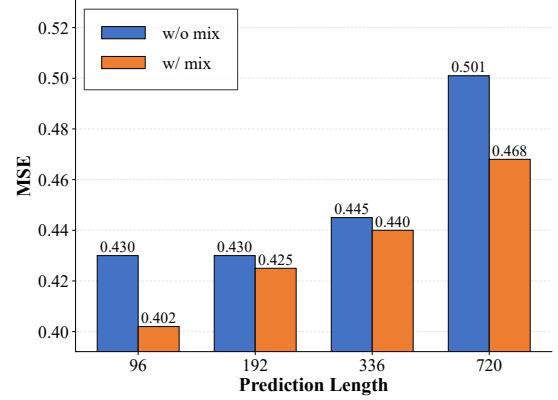


Fig. 7. MSE comparison on the Traffic dataset under different prediction lengths (96, 192, 336, and 720).

E. Visualization of Learned Representations

To intuitively understand how DACNet aligns local contexts with global patterns, we visualize the learned hidden representations and the corresponding attention heatmaps on the ETTm1 dataset (Period $P = 96$). We specifically analyze the Attention Heatmaps to decipher how the model queries the global memory. In these maps, the horizontal axis (Columns) represents the latent embedding dimension of the inter-cycle prototypes (i.e., the compressed global patterns), while the vertical axis (Rows) corresponds to the sequence of query samples.

1) Intra-Cycle Phase Sensitivity (Continuous Case): We first examine samples sampled continuously within a single cycle ($t \in \{1, 21, 41, 61\}$), as shown in Fig. 8a and Fig. 8b.

• **Representation Diversity:** In Fig. 8a, the extracted global features $\mathbf{h}^{(\text{glob})}$ exhibit significant morphological diversity. This confirms that even within the same global context, the model retrieves different feature shapes to match the evolving local phases.

• **Attention Stability:** Crucially, the attention heatmaps in Fig. 8b exhibit smooth and continuous vertical bands. This indicates that adjacent samples within a cycle tend to focus on the same or adjacent latent inter-cycle prototypes (columns). It implies that the “global trend context” remains stable within

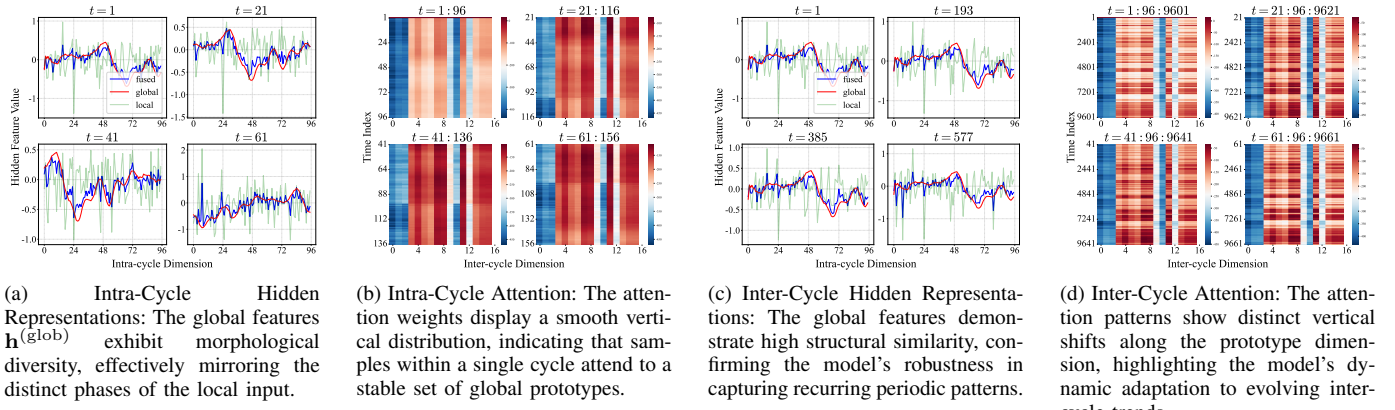


Fig. 8. Visualization of learned representations and attention patterns on ETTm1. (a) and (b) present the Intra-Cycle Analysis using continuous samples ($t \in \{1, 21, 41, 61\}$), while (c) and (d) present the Inter-Cycle Analysis using strided samples across different cycles ($t \in \{1, 193, 385, 577\}$).

a short time window, allowing the model to focus on phase variations without switching its global reference.

2) Inter-Cycle Trend Adaptation (Strided Case): Next, we analyze samples taken from the same relative phase but across distinct cycles ($t \in \{1, 193, 385, 577\}$), as presented in Fig. 8c and Fig. 8d.

- **Representation Similarity:** In Fig. 8c, the $h^{(\text{glob})}$ curves show high structural similarity, confirming the model correctly identifies the recurring periodic shapes.

- **Attention Shifting:** However, the attention heatmaps in Fig. 8d reveal a striking phenomenon: the high-attention regions jump abruptly between different columns along the vertical axis.

- **Insight:** This “jumping” signifies that for each new cycle, the model adaptively shifts its focus to a different latent prototype on the x-axis. Even though the samples share the same phase, they belong to different historical periods with distinct trends. The model effectively “selects” the specific inter-cycle prototype that best matches the current cycle’s trend intensity, thereby achieving precise trend adaptation.

V. CONCLUSION

We presented DACNet, an ultra-lightweight forecasting framework that distills global temporal patterns from the full historical training series. DACNet restructures the 1D history into a cyclemap and applies efficient compression and denoising to obtain a compact global representation, which is then dynamically aligned with each input window. By fusing the aligned global patterns with local features, DACNet captures complementary information across multiple temporal scales while maintaining a small parameter budget. Extensive experiments on eight real-world benchmarks demonstrate state-of-the-art accuracy with substantially improved efficiency.

Limitations: Despite its promising performance, we acknowledge certain limitations inherent to our design:

- **Periodicity Dependence:** The model relies on discernible periodic patterns, potentially limiting generalization to chaotic or aperiodic data where a stable period is absent.

- **Metadata Requirement:** Unlike models using only relative values, our alignment mechanism necessitates explicit temporal metadata (timestamps) during inference.

- **Static Nature:** The cyclemap is static post-training, which may cause adaptation lag under significant concept drift or sudden distribution shifts.

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